RAGFailBench revision: tables and figure

Model evaluation (Tables XII, XIII, XV, XVI)

Table 1: Score by condition, mean ± std over three seeded runs. Score is answer accuracy for answer-expected conditions (conflict: handled rate) and correct-abstention rate for abstain-expected ones; closed-book is accuracy with no context (prior knowledge, a probe outside FRS). Δ is the paired drop in percentage points against each item's own parent clean seed; FRS = 1 − mean paired drop.

Condition	nova-pro score (Δ pp)	llama score (Δ pp)	gpt-oss score (Δ pp)
clean	0.98 ± 0.01	0.99 ± 0.01	0.98 ± 0.01
missing_evidence (abstain)	0.92 ± 0.02 (-8.0)	0.93 ± 0.02 (-6.9)	0.93 ± 0.01 (-7.2)
hard_negative (abstain)	0.98 ± 0.01 (-2.3)	0.98 ± 0.01 (-2.1)	0.97 ± 0.01 (-2.6)
conflict	0.70 ± 0.03 (-28.5)	0.72 ± 0.02 (-26.6)	0.68 ± 0.02 (-30.4)
context_noise	0.98 ± 0.01 (-0.1)	0.99 ± 0.00 (+0.1)	0.99 ± 0.01 (+0.3)
chunk_boundary	0.96 ± 0.01 (-1.8)	0.97 ± 0.01 (-1.3)	0.93 ± 0.00 (-5.4)
evidence_position	0.98 ± 0.00 (+0.1)	0.99 ± 0.00 (+0.1)	0.98 ± 0.01 (-0.1)
closed-book (probe)	0.28 ± 0.03	0.47 ± 0.02	0.49 ± 0.03
FRS	0.932 ± 0.003	0.939 ± 0.008	0.924 ± 0.006

Table 2: Abstention / answered-correctly / hallucination rate by condition, mean ± std over three seeded runs. For abstain-expected operators a non-abstention is either correct (prior knowledge or a residual partial leak) or a hallucination (wrong answer).

Condition	nova-pro abst. / ans.-correct / halluc.	llama abst. / ans.-correct / halluc.	gpt-oss abst. / ans.-correct / halluc.
missing_evidence	0.92 ± 0.02 / 0.01 ± 0.01 / 0.07 ± 0.01	0.93 ± 0.02 / 0.01 ± 0.00 / 0.06 ± 0.01	0.93 ± 0.01 / 0.01 ± 0.01 / 0.06 ± 0.01
hard_negative	0.98 ± 0.01 / 0.00 ± 0.00 / 0.02 ± 0.01	0.98 ± 0.01 / 0.00 ± 0.00 / 0.02 ± 0.01	0.97 ± 0.01 / 0.00 ± 0.00 / 0.02 ± 0.01
conflict	0.01 ± 0.00 / — / 0.30 ± 0.03	0.01 ± 0.01 / — / 0.27 ± 0.01	0.01 ± 0.00 / — / 0.31 ± 0.02
context_noise	0.00 ± 0.00 / — / 0.02 ± 0.00	0.00 ± 0.00 / — / 0.01 ± 0.01	0.00 ± 0.00 / — / 0.01 ± 0.00
chunk_boundary	0.02 ± 0.00 / — / 0.02 ± 0.01	0.01 ± 0.01 / — / 0.01 ± 0.01	0.06 ± 0.00 / — / 0.02 ± 0.00
evidence_position	0.00 ± 0.00 / — / 0.02 ± 0.00	0.00 ± 0.00 / — / 0.01 ± 0.00	0.00 ± 0.00 / — / 0.01 ± 0.00

Table 3: Paired drop (pp) by severity: mean ± std over three seeded runs of the three-model mean. For abstain-expected operators the drop is the rate of failing to abstain; missing_evidence severity is closeness of the remaining context to the removed evidence.

Operator	low	medium	high
missing_evidence	0.0 ± 0.0	7.8 ± 2.3	15.7 ± 2.6
hard_negative	1.7 ± 1.3	2.2 ± 0.5	3.1 ± 0.9
conflict	12.2 ± 1.7	28.9 ± 2.0	44.4 ± 2.1
context_noise	-0.1 ± 0.6	-0.1 ± 0.8	-0.1 ± 0.8
chunk_boundary	0.4 ± 0.7	3.0 ± 0.3	5.8 ± 1.2
evidence_position	-0.2 ± 0.6	0.1 ± 0.6	-0.1 ± 0.9

Table 4: Conflict outcomes, mean ± std (%) over three seeded runs. Handled = gold, both values, or an acknowledged conflict. Aware = prompt that permits flagging a conflict (probe, excluded from FRS). Prior known/unknown splits by closed-book correctness.

Model	Prompt	Gold %	Alternate %	Both %	Acknowledged %	Handled %	Judge-only %	Handled, prior known %	Handled, prior unknown %
nova-pro	default	59.3 ± 2.4	25.8 ± 4.2	10.4 ± 1.3	10.6 ± 1.5	69.7 ± 3.4	69.2 ± 3.2	86.5 ± 4.0	63.2 ± 3.0
nova-pro	aware	30.5 ± 1.7	9.3 ± 1.6	55.8 ± 2.1	58.5 ± 2.0	88.6 ± 0.8	37.1 ± 1.2	95.0 ± 2.8	86.1 ± 0.1
llama	default	36.8 ± 1.9	18.2 ± 1.6	34.9 ± 1.2	35.1 ± 1.4	72.0 ± 2.3	62.1 ± 2.8	78.0 ± 3.7	66.6 ± 4.3
llama	aware	13.7 ± 0.4	2.3 ± 0.3	77.1 ± 0.8	83.9 ± 0.2	95.5 ± 0.3	31.3 ± 2.9	97.5 ± 0.2	93.8 ± 0.6
gpt-oss	default	50.4 ± 1.9	25.2 ± 2.2	17.5 ± 2.5	17.6 ± 2.7	67.9 ± 2.1	64.2 ± 2.6	78.2 ± 3.0	58.1 ± 3.9
gpt-oss	aware	13.6 ± 2.5	2.6 ± 0.4	74.5 ± 5.7	84.9 ± 3.2	95.6 ± 0.8	23.2 ± 3.4	95.5 ± 1.1	95.7 ± 1.1

Table 5: Paired drop (pp) and FRS with 95% CIs, pooled over three seeded runs; cluster bootstrap over 561 parent clean seeds, 2000 resamples.

Condition	nova-pro	llama	gpt-oss
missing_evidence	8.0 [6.6, 9.6]	6.8 [5.4, 8.3]	7.2 [5.8, 8.7]
hard_negative	2.3 [1.3, 3.4]	2.1 [1.3, 3.1]	2.6 [1.5, 3.8]
conflict	28.5 [25.6, 31.4]	26.6 [24.1, 29.3]	30.4 [27.6, 33.3]
context_noise	0.1 [-0.9, 1.0]	-0.1 [-0.9, 0.7]	-0.3 [-1.0, 0.4]
chunk_boundary	1.8 [0.7, 2.8]	1.2 [0.2, 2.4]	5.4 [3.8, 7.0]
evidence_position	-0.1 [-1.1, 0.8]	-0.1 [-0.7, 0.4]	0.1 [-0.6, 0.8]
FRS	0.932 [0.925, 0.940]	0.939 [0.933, 0.945]	0.924 [0.917, 0.931]

Table 6: FRS difference between models with paired 95% CIs (same resampled seeds), pooled over three seeded runs; cluster bootstrap over 561 parent clean seeds, 2000 resamples.

Models	ΔFRS [95% CI]	CI excludes 0
nova-pro – llama	-0.006 [-0.015, +0.002]	no
nova-pro – gpt-oss	+0.008 [-0.000, +0.016]	no
llama – gpt-oss	+0.015 [+0.006, +0.022]	yes

Confidence intervals, McNemar tests, correctness signals (Tables XVII, XIX)

Table 7: Exact McNemar tests between models on the same items (expected-behavior score; pooled over three seeded runs). Discordant = items only the first / only the second model gets right; p (Holm) corrects over all 21 tests.

Condition	Models	n	Δ pp	Discordant	p	p (Holm)
clean	nova-pro vs llama	561	-0.4	4 / 6	0.75	1
clean	nova-pro vs gpt-oss	561	+0.0	5 / 5	1	1
clean	llama vs gpt-oss	561	+0.4	4 / 2	0.69	1
missing_evidence	nova-pro vs llama	1552	-1.2	28 / 46	0.047	0.85
missing_evidence	nova-pro vs gpt-oss	1552	-0.8	32 / 44	0.21	1
missing_evidence	llama vs gpt-oss	1552	+0.4	37 / 31	0.54	1
hard_negative	nova-pro vs llama	1605	-0.2	13 / 16	0.71	1
hard_negative	nova-pro vs gpt-oss	1605	+0.3	16 / 11	0.44	1
hard_negative	llama vs gpt-oss	1605	+0.5	21 / 13	0.23	1
conflict	nova-pro vs llama	1683	-2.3	188 / 226	0.069	1
conflict	nova-pro vs gpt-oss	1683	+1.8	235 / 204	0.15	1
conflict	llama vs gpt-oss	1683	+4.1	241 / 172	0.0008	0.015
context_noise	nova-pro vs llama	1683	-0.5	17 / 25	0.28	1
context_noise	nova-pro vs gpt-oss	1683	-0.4	15 / 21	0.41	1
context_noise	llama vs gpt-oss	1683	+0.1	17 / 15	0.86	1
chunk_boundary	nova-pro vs llama	1368	-0.8	15 / 26	0.12	1
chunk_boundary	nova-pro vs gpt-oss	1368	+3.8	70 / 18	2.1e-08	4.2e-07
chunk_boundary	llama vs gpt-oss	1368	+4.6	73 / 10	5.8e-13	1.2e-11
evidence_position	nova-pro vs llama	1683	-0.4	15 / 21	0.41	1
evidence_position	nova-pro vs gpt-oss	1683	+0.2	22 / 18	0.64	1
evidence_position	llama vs gpt-oss	1683	+0.6	18 / 8	0.076	1

Table 8: Share of answer-expected items each correctness signal accepts, pooled over three seeded runs. The reported answer accuracy is the union (exact match, token-F1 ≥ 0.6, or judge). Conflict here is answer correctness before outcome scoring.

Condition	Model	Exact match	Token-F1 ≥ 0.6	Judge	Union (reported)
clean	nova-pro	0.731	0.879	0.971	0.982
clean	llama	0.745	0.884	0.975	0.986
clean	gpt-oss	0.688	0.843	0.973	0.982
conflict	nova-pro	0.434	0.603	0.595	0.692
conflict	llama	0.219	0.497	0.442	0.621
conflict	gpt-oss	0.295	0.516	0.504	0.642
context_noise	nova-pro	0.730	0.875	0.970	0.982
context_noise	llama	0.747	0.887	0.976	0.986
context_noise	gpt-oss	0.687	0.847	0.976	0.985
chunk_boundary	nova-pro	0.751	0.874	0.962	0.965
chunk_boundary	llama	0.773	0.902	0.969	0.973
chunk_boundary	gpt-oss	0.689	0.819	0.925	0.927
evidence_position	nova-pro	0.736	0.877	0.973	0.983
evidence_position	llama	0.747	0.891	0.977	0.987
evidence_position	gpt-oss	0.693	0.853	0.973	0.981

Table 9: Conflict paired drop (pp) / FRS under three conflict scoring rules, pooled over three seeded runs. Handled = gold, both values, or an acknowledged conflict; gold only = commits to the gold value; answer correctness = exact match, token-F1 ≥ 0.6 or judge, with no credit for naming both. The model ranking on FRS depends on the rule.

Conflict rule	nova-pro	llama	gpt-oss	FRS ranking
Handled (reported)	28.5 / 0.932	26.6 / 0.939	30.4 / 0.924	llama > nova-pro > gpt-oss
Gold only	38.9 / 0.915	61.7 / 0.880	47.8 / 0.895	nova-pro > gpt-oss > llama
Answer correctness	29.1 / 0.932	36.5 / 0.923	34.0 / 0.918	nova-pro > llama > gpt-oss

Table 10: Human audit summary (replaces the HAR and failure-validity rows of Table V). Blind stratified sample from the three seeded runs; 95% Wilson CIs. Annotator 1 is primary.

Metric	Annotator 1	Annotator 2
Human acceptance (HAR), clean seeds	84.0% [71.5, 91.7] (42/50)	86.0% [73.8, 93.0] (43/50)
Failure injection valid	96.3% [87.5, 99.0] (52/54)	98.1% [90.2, 99.7] (53/54)
Failure label (answer available) correct	100.0% [93.4, 100.0] (54/54)	100.0% [93.4, 100.0] (54/54)

Table 11: Human audit per seeded run (replaces Tables VI and VIII human columns): clean seeds kept and failure cases judged valid, per annotator. Per-run samples are small (16–17 seeds, 18 failure cases); pooled values are in the summary table.

Run	HAR A1	HAR A2	Failure validity A1	Failure validity A2
pilot_stability_s42	16/16	16/16	17/18	17/18
pilot_stability_s123	13/17	13/17	17/18	18/18
pilot_stability_s2026	13/17	14/17	18/18	18/18
All runs	42/50	43/50	52/54	53/54

Human audit (Tables V, VI, VIII, IX, XI human rows)

Table 12: Primary reason for the 8 of 50 audited clean seeds annotator 1 did not keep (replaces Table IX). One primary reason per row from the structured flags, in the order listed; the free-text note gives the detail.

Reason	Count	Percent	Structured basis	Rows
Time-sensitive phrasing	2	25.0%	time_sensitive_ok = no	C022 (fix), C027 (fix)
Answer not unique	1	12.5%	answer_unique = no	C002 (fix)
Unclear question (reference, premise, wording)	4	50.0%	question_clear = no	C010 (fix), C039 (fix), C044 (fix), C047 (fix)
Other (trivial; see note)	1	12.5%	no flag set	C040 (reject)

Table 13: Failure-injection verification: automatic check on every case versus the human audit (replaces Table XI). Per-operator samples are 9 cases (3 per severity).

Check	Annotator 1	Annotator 2
Automatic structural verification (leakage / label check), all cases	94.9% ± 0.3% pass	
Human: injection valid, all operators	96.3% [87.5, 99.0] (52/54)	98.1% [90.2, 99.7] (53/54)
missing_evidence	9/9	9/9
hard_negative	9/9	9/9
conflict	7/9	8/9
context_noise	9/9	9/9
chunk_boundary	9/9	9/9
evidence_position	9/9	9/9

Table 14: Annotator 1 vs annotator 2 on the same audit rows. Annotator 2 had seen annotator 1's labels in an earlier pass, so this is a consistency check, not inter-annotator reliability. Kappa is undefined when both used a single identical label.

Field	n	Agreement	Cohen's kappa
decision	50	98.0%	0.92
question_clear	50	98.0%	0.85
answer_in_evidence	50	100.0%	undefined (one label only)
answer_unique	50	98.0%	0.66
time_sensitive_ok	50	100.0%	1.00
needs_title	50	100.0%	1.00
human_injection_valid	54	98.1%	0.66
human_label_correct	54	100.0%	undefined (one label only)
severity_ok	54	74.1%	0.14

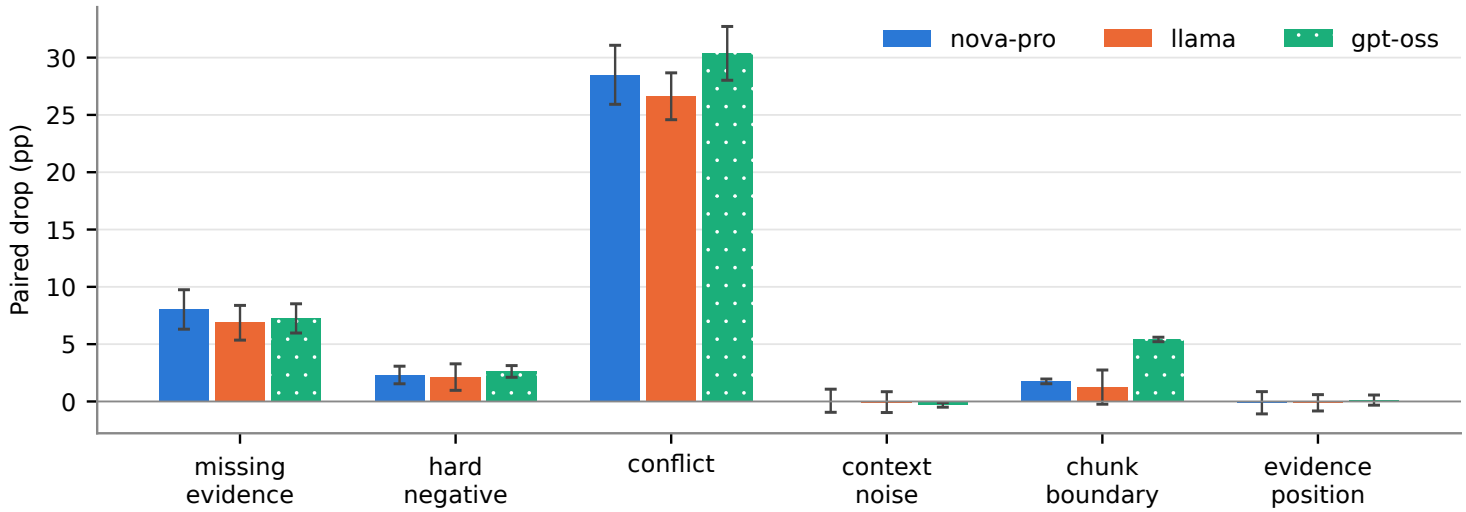


Figure 1: Paired performance drop by failure operator (Fig. 2).